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ID publico: case_0222

Paginas del documento: 5

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semana después. Los datos se compararon en el pre y postoperatorio mediante la prueba t de Student para *sto*-efectividad y el valor necesario para mejorar un punto en la EVA en relación con la edad se calcularon *isión* y regla de tres. Para el análisis inferencial, se utilizó la prueba t de Welch emparejada con Bootstrap y s. Resultados: En el dolor lumbar, se observó una variación de 2 a 10, con una media de $7,5 \pm 2,14$ (IC95%: 6,8-8,1) antes del procedimiento y una variación de 0 a 10, con una media de $3,1 \pm 2,8$ (IC95%: 2,3-4,0) una semana después ($p < 0,0001$). En el dolor en las piernas, se observó una variación de 1 a 10, con una media de $6,8 \pm 2,5$ (IC95%: 6,1-7,4) antes de la intervención y una variación de 0 a 9, con una media de $2,4 \pm 2,5$ (IC95%: 1,8-3,1) una semana después ($p < 0,0001$). El costo de los materiales utilizados durante cada procedimiento fue de 214,72 reales. Los pacientes mayores de 60 años presentaron una mejor costo-efectividad en comparación con los pacientes más jóvenes en infiltraciones lumbares, con la prueba T de Welch con Bootstrap $56,0$ [21,5;90,2] ($p = 0,006$). En relación al dolor en las piernas, no hubo diferencia significativa, con la prueba T de Welch con Bootstrap $-32,4$ [-70,9;7,00] ($p = 0,12$). Conclusión: La combinación de los procedimientos de bloqueo epidural sacral y bloqueo foraminal en pacientes mayores de 60 años para el dolor lumbar mostró una mejor costo-efectividad, mientras que, en relación al dolor en las piernas, no hubo diferencia significativa en la costo-efectividad del procedimiento. **Nivel de Evidencia III; Estudio de Cohorte Retrospectiva.**

Descriptor: Bloqueo Nervioso; Enfermedades Crónicas; Dolor de la Región Lumbar; Dolor; Grupos de Edad.

INTRODUCTION

Lower back pain is one of the leading causes of disability worldwide, affecting individuals of all ages, but with the highest prevalence between 35 and 55 years of age.¹ The etiology is most often multifactorial, involving degenerative changes associated with aging, as well as fractures, tumors, and infections.¹ Genetic, biomechanical and environmental factors also play significant roles in the development of chronic back pain.^{1,2}

The symptoms of degenerative spine diseases can occur in up to 65% of the world population annually and up to 84% of people at some point in life,^{3,4} the most common being lumbar pain and radiating pain to the legs.⁴ It is estimated that 5 to 10% of individuals who experience lumbar pain will develop chronic pain. When correlated with age, the prevalence of lumbar pain and radiated pain to the lower limbs is 3 to 4 times higher in individuals over 50 years of age compared to those between the ages of 18 and 30 years.^{3,4} In addition, women, people with lower income, lower education, and smokers presented a higher prevalence of chronic lumbar pain and radiated pain.^{5,6}

Several authors have demonstrated an association between the symptoms resulting from these changes and the reduction in the patient's quality of life.^{7,8} Such symptomatology is responsible for frequent departures to early care, medical leave, and individual suffering, thereby generating high treatment costs.⁹ Thus, the treatment aims at improving the patient's clinical conditions and, consequently, improving their quality of life.^{9,10}

Treatments for lumbar pain involve both surgical and non-surgical measures, some of which include physical therapies and rehabilitation, pharmacotherapy, and behavioral therapies.¹¹ The surgical options include analgesic blocks, microdiscectomies, lumbar fusion (arthrodesis), arthroplasties, decompressions, among others.^{12,13} The decision on surgical or conservative treatment will depend on the symptoms and their severity; surgery is indicated in severe cases, such as loss of strength, instability, or nerve compressions that do not respond well to conservative treatment.¹⁴⁻¹⁶

Facet blocks and epidural blocks are gaining prominence in the treatment of symptoms arising from degenerative diseases of the spine.^{17,18} The injection of medications directly into the spine with the help of an image enhancer is a minimally invasive procedure aimed at confirming the presumptive diagnosis as well as improving pain.¹⁹ For this purpose, various materials are currently available, ranging from spinal anesthesia needles to radiofrequency cannulas and specific cannulas designed for performing spinal blocks.^{20,21}

Given this, the present study aimed to identify the relationship between the cost-effectiveness of facet and sacral epidural blocks

MATERIALS AND METHODS

Ethical aspects

The study was approved by the Research Ethics Committee of the institution where it was conducted (CAAE: 48835721.8.0000.8098), by the principles outlined in Resolution No. 466/2012 of the National Health Council regarding the parameters of research involving humans. This is an observational study, specifically a retrospective cohort, employing both quantitative and qualitative analysis approaches. Because it was a secondary study in which patients were not directly investigated, a waiver of the need for formalized processing, as outlined in the Terms of Free and Informed Consent, was requested and granted.

Sample characteristics

In this study, 47 clinical studies were evaluated in which patients presented with lumbar pain associated with radiating pain to the lower limbs and were refractory to non-surgical treatment for more than four weeks. All patients included in this study were treated at the same clinic specializing in the treatment of spine diseases, and all procedures were performed at the same hospital by the same professional, who has more than 10 years of experience in spine surgery.

We considered as the inclusion criterion individuals of both genders, aged over 18 years, with lumbar pain and radiated pain to the refractory leg, who had undergone conservative treatment with analgesic medication and physiotherapy for more than four weeks, and who were treated through a combination of sacral epidural and foraminal blocks. The exclusion criteria included patients under 18 years of age, a diagnosis of tumor disease, infection, trauma to the spine, and a history of previous lumbar surgery. For confirmation of the diagnosis, complementary imaging methods, including radiography and magnetic resonance imaging, were used in all cases.

Evaluated variables

Among the information obtained from the medical records, we can cite the age of patients, gender, the total cost of materials used during the procedure, and the intensity of pain as measured by the visual analog scale before and after a week of infiltrations. All this information was obtained from the records present in the clinic and the hospital where the studied procedures were performed.

The total cost of the procedure was calculated by summing the costs of all devices, materials, and medicines used. All this information was obtained from the hospital's records, where the

nce between VASi and VASf was less
X was arbitrarily adjusted for the total

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years. The age of female patients ranged from 26 to 84 years, with an average of 60.13 years; and the age of male patients ranged from 24 to 86 years, with an average of 49.17 years. Among the diagnoses of the patients evaluated, we had: facet degeneration, degeneration of the intervertebral disc, vertebral canal stenosis, disc hernia, and patients with grade 1 vertebral listhesis. Regarding the complications, no occurrences were identified.

Concerning lumbar pain, values were identified that ranged from 2 to 10, with an average of 7.5 ± 2.14 (95% CI 6.8 – 8.1) before the procedure (VASi) and the average of 0 to 10, with an average of 3.1 ± 2.8 (95% CI 2.3 – 4.0) a week after ($p < 0.0001$) the procedure (VASf). (Table 1)

About pain in the lower limbs, values were identified that ranged from 1 to 10, with an average of 6.8 ± 2.5 (95% CI 6.1 – 7.4) before the procedure (VASi) and one varied from 0 to 9, with an average of 2.4 ± 2.5 (95% CI 1.8 – 3.1) a week after ($p < 0.0001$) the procedure (VASf). (Table 1)

The total value of the procedure was R\$ 214.75 and was the same for all patients included in the study. (Table 2)

Patients over 60 years of age showed better cost effectiveness (lower value required for 1 VAS improvement) compared to younger patients when receiving infiltrations for lumbar pain. (Table 3)

However, with regard to pain in the legs, there was no significant difference between the cost and effectiveness of the procedure among patients over 60 years of age compared to younger patients when receiving infiltrations for pain in the legs. (Table 4)

Table 1. Assessment of pain in the lumbar spine (n=45) and lower limbs (n=56) by visual analog scale (VAS) in patients undergoing epidural block.

Pain	Preoperative	Postoperative	p
Low Back	7.5 ± 2.14	3.1 ± 2.8	< 0.0001
	(95% CI 6.8 – 8.1)	(95% CI 2.3 – 4.0)	
Lower members	6.8 ± 2.5	2.4 ± 2.5	< 0.0001
	(95% CI 6.1 – 7.4)	(95% CI 1.8 – 3.1)	

Table 2. Materials used to perform sacral epidural and foraminal blocks, with their respective quantities and costs in real (Adapted Authors).

Material used	Quantity	Unit value (R\$)	Total value (R\$)
Alfentanil 0.5 mg/ml	1 ampoule	14.99	14.99
Midazolam 1mg/ml	1 ampoule	5.27	5.27
Ringer lactate 500 ml	1 bag	2.14	2.14
Abocath 24G (0.7x119 mm)	1 unit	2.53	2.53
Nasal oxygen catheter	1 unit	0.91	0.91
Microdrip infusion set	1 unit	3.82	3.82
Tegaderm peripheral catheter	1 unit	5.45	5.45
Needle 25mm x 8mm	1 unit	0.1	0.1
Spinal needle	3 units	13.56	40.68
10 ml syringe with thread	2 units	0.5	1
5 ml syringe with thread	1 unit	0.33	0.33
5 ml syringe without thread	1 unit	0.33	0.33
Betamethasone dipropionate + disodium phosphate 5 mg +2 mg	1 unit	3.25	3.25
Non-ionic iodized contrast 300 mg/ml	1 unit	34.68	34.68
Lidocaine 2%	1 unit	6.11	6.11
Triamcinolone 20mg/ml	1 unit	81.2	81.2
Sterile operating field	1 unit	7.06	7.06
Chlorhexidine 0.5% alcoholic solution 100 ml	1 unit	1.87	1.87
Sterile Gaze	2 units	0.7	1.4
Surgical glove	1 unit	1.63	1.63

Statistical analysis

The data were tabulated using Microsoft Excel 2016 software. The data obtained were analyzed statistically using the Stata software, using averages, standard deviations, and a 95% confidence interval.

For inferential analysis, the Welch t-test, paired with Bootstrap and the robust median test, was used to determine whether there is a difference in the cost and effectiveness of treatment among

				Bootstrap
		/179.7	18.0/179.7	56.0 [21.5;90.2]. p= 0.006
				Robust Median Difference Test
				44.9 [0; 154.02]. p=0.043
	Med [IQI]	74.9 [31.4;179.7]	29.9 [25.7;44.9]	
	Average (SD)	101.5 (74.6)	45.3 (42.3)	
	N (NA)	22 (0)	25 (0)	

Legend: Med: Median; IQI: Interquartile interval; SD: Standard deviation; NA: Missing or Not Applicable; N = Number; CI: Confidence interval.

Table 4. Comparison of the cost to improve 1 VAS leg between age groups.

Variable	Statistics	Age group		Test Effect [95% CI]
		59-	60 +	
				Welch T-Test with Bootstrap
Cost by improvement of 1 VAS	Min/Max	18.0/179.7	20.0/179.7	-32.4 [-70.9;7.00]. p=0.12
				Robust Median Difference Test
				-119 [-149.4;0]. p=0.29
	Med [IQI]	59.9 [25.7;179.7]	179.7 [44.9;179.7]	
	Average (SD)	96.9 (75.0)	129.7 (68.7)	
	N (NA)	21 (0)	33 (0)	

Legend: Med: Median; IQI: Interquartile interval; SD: Standard deviation; NA: Missing or Not Applicable; N = Number; CI: Confidence interval.

DISCUSSION

When comparing the cost-effectiveness of the combination of foraminal and sacral blocks across different patient age groups, this study demonstrated that, regarding lumbar spine pain, patients over 60 years old exhibited better cost-effectiveness compared to younger patients. However, no statistically significant difference was found concerning pain in the lower limbs. Additionally, the low cost of performing the procedure was also highlighted.

The use of corticosteroid blocks for the management of lumbar and sciatic pain has been studied and performed since 1953, when Lieve reported the first peridural injection of hydrocortisone.¹¹

CONCLUSION

Patients with lumbar pain over 60 years of age undergoing a combination of foraminal and sacral block presented better cost-effectiveness compared to younger patients in the study sample. Regarding the pain irradiated to the lower limbs, there was no significant difference in cost and effectiveness between patients over 60 years of age and those who were younger.

All authors declare no potential conflict of interest related to this article.

CONTRIBUTIONS OF THE AUTHORS: Each author contributed individually and significantly to the development of this article. NMJ: conceptualization, data curation, original draft writing; LCCA: formal analysis, research and validation; MA: project administration, supervision and visualization; PMD.: meteorology; HCFPS.: validation and writing- review and editing.

REFERENCES

